

Project-I Proposal Report
On
One-Stop Personalized Career & Education Advisor

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In partial fulfillment of requirement for the degree of
Bachelor of Technology



Jodhpur Institute of Engineering & Technology

(An Autonomous Institution affiliated with Bikaner Technical University, Bikaner)

Date of Submission:

December 26, 2025

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Chapter 1

Introduction

In India, a significant number of students face uncertainty and confusion while making academic and career decisions after completing Class 10 and Class 12. Despite the availability of adequate infrastructure and qualified faculty in government degree colleges, enrollment in these institutions has been steadily declining. This challenge is not primarily due to issues related to quality, but rather due to limited awareness about the importance of higher education and the long-term career opportunities associated with different degree programs.

A major concern lies in the absence of structured and personalized career guidance at the school level. Most students and their parents have limited understanding of subject streams such as Arts, Science, and Commerce, and the academic and professional pathways each stream enables. As a result, students often make uninformed decisions, choose unsuitable courses, or opt for short-term skill-based programs without considering long-term academic growth and career stability.

Another critical issue is the limited access to reliable and consolidated information about nearby government colleges. Details regarding available degree programs, eligibility criteria, admission timelines, facilities, and future prospects are often scattered across multiple sources or difficult to access, particularly for students from rural and economically weaker backgrounds. This lack of localized guidance contributes to confusion, higher dropout rates, and migration toward private institutions that may not always meet educational expectations.

With advancements in digital technology and artificial intelligence, there is a growing opportunity to address these challenges through personalized digital guidance platforms. Such systems can assess student aptitude and interests, provide data-driven recommendations, and visually map academic courses to real-world career opportunities. These capabilities can significantly enhance clarity and confidence in academic decision-making.

The proposed project, titled **“One-Stop Personalized Career & Education Advisor”**, aims to develop a comprehensive digital platform that offers aptitude-based course recommendations, course-to-career mapping, nearby government college information, and admission timeline tracking. The platform is designed to support students, parents, and educators by delivering structured, localized, and personalized career guidance.

By integrating career awareness into the academic decision-making process, this project seeks to improve enrollment in government degree colleges, reduce student dropouts, and empower learners with clarity and confidence regarding their academic and professional futures. The proposed solution aligns with national education goals and supports the broader vision of using technology to create inclusive, informed, and sustainable education systems.

Chapter 2

Problem Statement

Students in India, particularly those completing Class 10 and Class 12, often face significant challenges in making informed academic and career decisions. Despite the availability of government degree colleges with adequate infrastructure and qualified faculty, enrollment in these institutions continues to decline. This issue primarily arises from limited awareness regarding the importance of graduation, the variety of available degree programs, and the long-term career opportunities associated with different academic streams.

Career guidance for students is largely unstructured and inconsistent across schools and regions. Many students and parents have limited understanding of subject streams such as Arts, Science, and Commerce, and the academic as well as professional pathways enabled by each option. As a result, students frequently make uninformed choices, select unsuitable courses, or pursue short-term skill-based programs without considering long-term academic development and career stability.

Another major concern is the absence of a centralized and accessible source of information regarding nearby government colleges. Information related to degree programs, eligibility criteria, admission timelines, cut-offs, facilities, and future prospects is often scattered across multiple platforms or difficult to access, particularly for students from rural and economically weaker backgrounds. This lack of clarity leads to confusion, higher dropout rates, and migration to private institutions that may not provide the expected educational outcomes.

The significance of these challenges has been formally recognized at the national level through initiatives such as the **Smart India Hackathon (SIH)**, which has identified the need for a unified digital career and education guidance platform to support students in making informed academic and career decisions aligned with government education objectives [1].

Existing career counseling initiatives are mostly manual, limited in reach, and lack personalization.

Students often do not receive guidance tailored to their aptitude, interests, or regional educational opportunities. There is a clear need for a comprehensive digital guidance system that can assess student aptitude, map academic courses to career opportunities, provide localized college information, and deliver timely updates related to admissions and scholarships.

The problem addressed in this project is to design and develop a **One-Stop Personalized Career & Education Advisor** that leverages digital technologies and data-driven recommendations to support students in making informed academic and career choices. The proposed system aims to bridge the gap between academic awareness and practical decision-making, particularly for students aspiring to pursue higher education through government institutions.

2.1 Project Objectives

The main objectives of the proposed project are as follows:

- To develop a digital platform that provides structured and personalized career and education guidance for students.
- To assess students' aptitude, interests, and strengths through interactive quizzes and evaluations.
- To recommend suitable subject streams and degree programs based on individual profiles.
- To provide detailed mapping between degree courses and career opportunities, including jobs, government examinations, and higher education options.
- To offer location-based information on nearby government colleges, including available programs, eligibility criteria, and facilities.
- To track admission schedules, scholarship deadlines, and counseling timelines through notifications.
- To empower students to make informed academic decisions and reduce dropouts after Class 10 and Class 12.

2.2 Project Scope

The scope of this project includes the design, development, and implementation of a mobile and web-based personalized career and education guidance platform. The system will provide aptitude-based

recommendations, course-to-career mappings, government college directories, and academic timeline tracking.

The platform will support personalized user profiles to ensure recommendations are aligned with students' interests, academic background, and regional context. It will be designed to be scalable, lightweight, and accessible, including support for users in areas with limited internet connectivity.

The project focuses on guidance, awareness, and decision-support mechanisms and does not replace formal academic counseling bodies or educational authorities. Future enhancements may include artificial intelligence-based predictive analytics, multilingual support, integration with government education portals, and advanced data visualization to further improve the accuracy and reach of career guidance.

Chapter 3

Literature Survey

Career guidance and educational decision-making have been widely researched due to their long-term impact on students' academic success, employability, and socio-economic mobility. Studies indicate that early and structured career guidance helps students make informed academic choices and reduces dropouts after key transition points such as Class 10 and Class 12. However, access to such guidance remains uneven, particularly for students from government schools and rural backgrounds.

With the growth of digital education technologies, several career guidance platforms have emerged. In India, the **National Career Service (NCS)** portal provides employment-oriented guidance but offers limited personalized academic planning. Similarly, private platforms such as **Mindler**, **iDream-Career**, **CareerGuide.com**, and **Univariety** demonstrate the effectiveness of aptitude-based recommendations but are often paid and focused on private institutions.

Globally used platforms such as **My Next Move**, **Career Explorer**, and **Indeed Career Guide** offer skill-based career exploration; however, they are not aligned with the Indian education system or government college structures. Learning platforms such as **Coursera**, **edX**, **Udemy**, and **SWAYAM** focus on content delivery rather than academic decision guidance.

Reports by organizations such as **UNESCO**, the **World Bank**, and the **OECD** emphasize the need for structured, personalized, and technology-enabled career guidance integrated into formal education systems.

3.1 User Survey Analysis (Google Form Responses)

To complement the literature review, a primary user survey was conducted using a Google Form to understand student needs, challenges, and expectations regarding career and education planning [6].

The survey results provide strong empirical support for the proposed system.

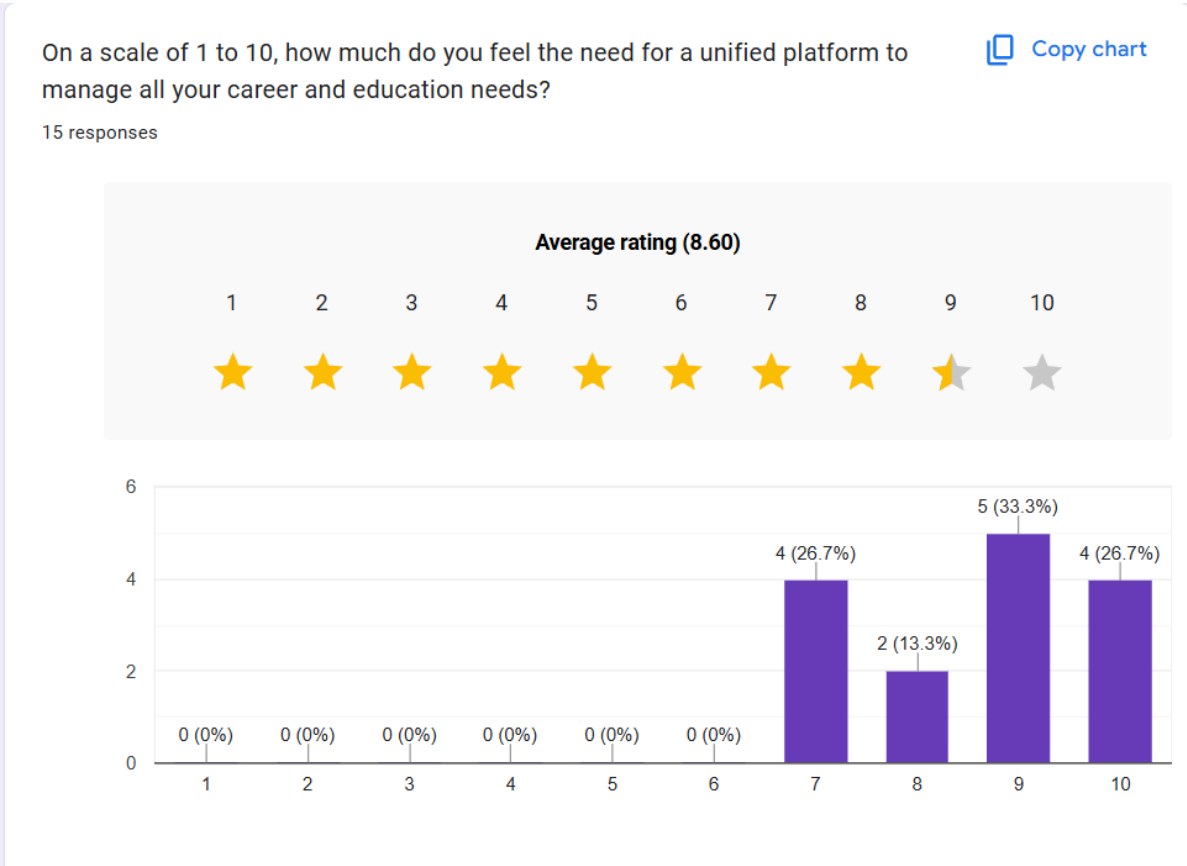


Figure 3.1: Perceived need for a unified career and education guidance platform

The survey indicates a strong demand for a centralized solution, with an average rating of **8.6 out of 10** for the need for a unified platform to manage career and education decisions.

Most respondents identified **AI-driven career path recommendations** as the most critical feature, followed by progress tracking tools and access to verified institutional and job data.

The most common frustrations reported include lack of progress, difficulty in finding reliable information, and repeated rework caused by unclear guidance.

Respondents showed high interest in curated course recommendations, real-time job market insights, and mentorship matching features.

A majority of respondents rely on online job portals, AI-powered tools, and informal sources such as family and friends, indicating the lack of a trusted centralized platform.

Most respondents reported moderate to high difficulty in gathering reliable information about career paths and educational programs.

The survey shows that students are actively exploring options or planning next steps, reinforcing the importance of timely and structured guidance.

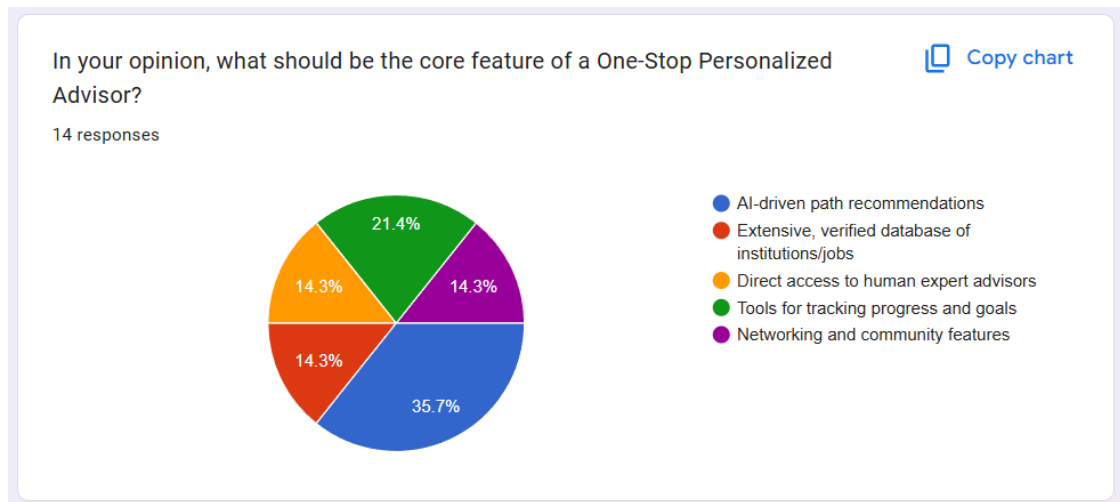


Figure 3.2: Preferred core features of a one-stop personalized advisor

Overall, the survey findings align strongly with existing research and clearly justify the need for a comprehensive, personalized, and accessible career guidance platform. The proposed **One-Stop Personalized Career & Education Advisor** directly addresses these gaps by integrating aptitude assessment, career mapping, government college discovery, and academic timeline tracking within a single system.

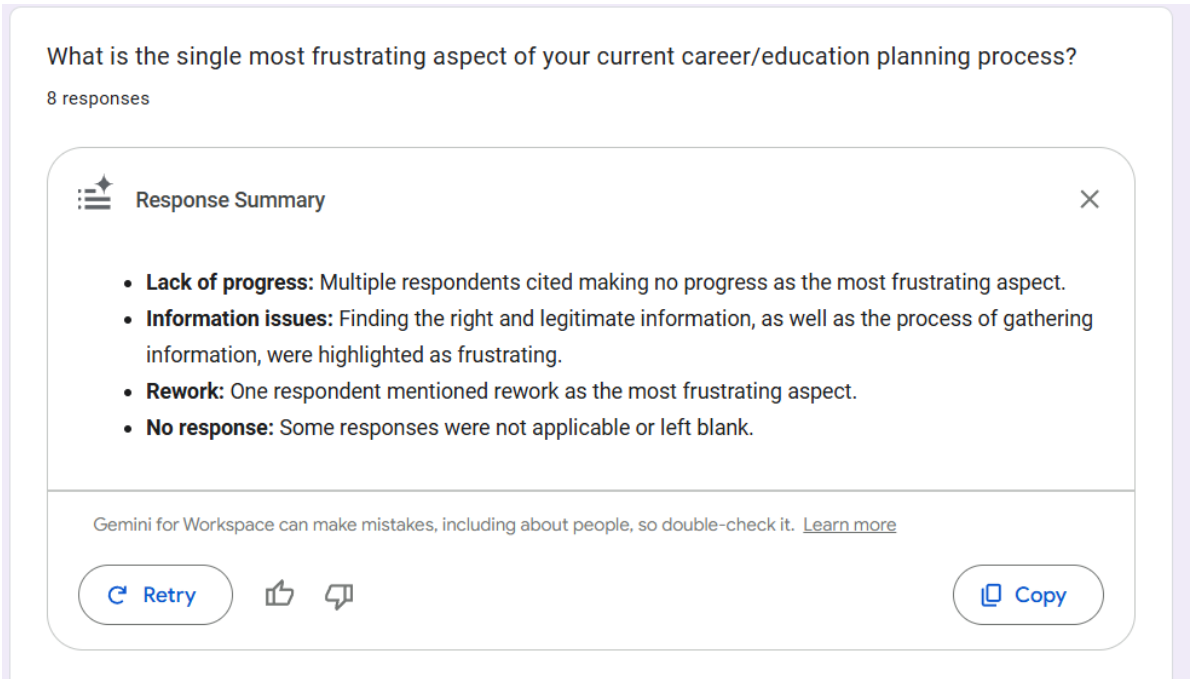


Figure 3.3: Most frustrating aspects of current career planning processes

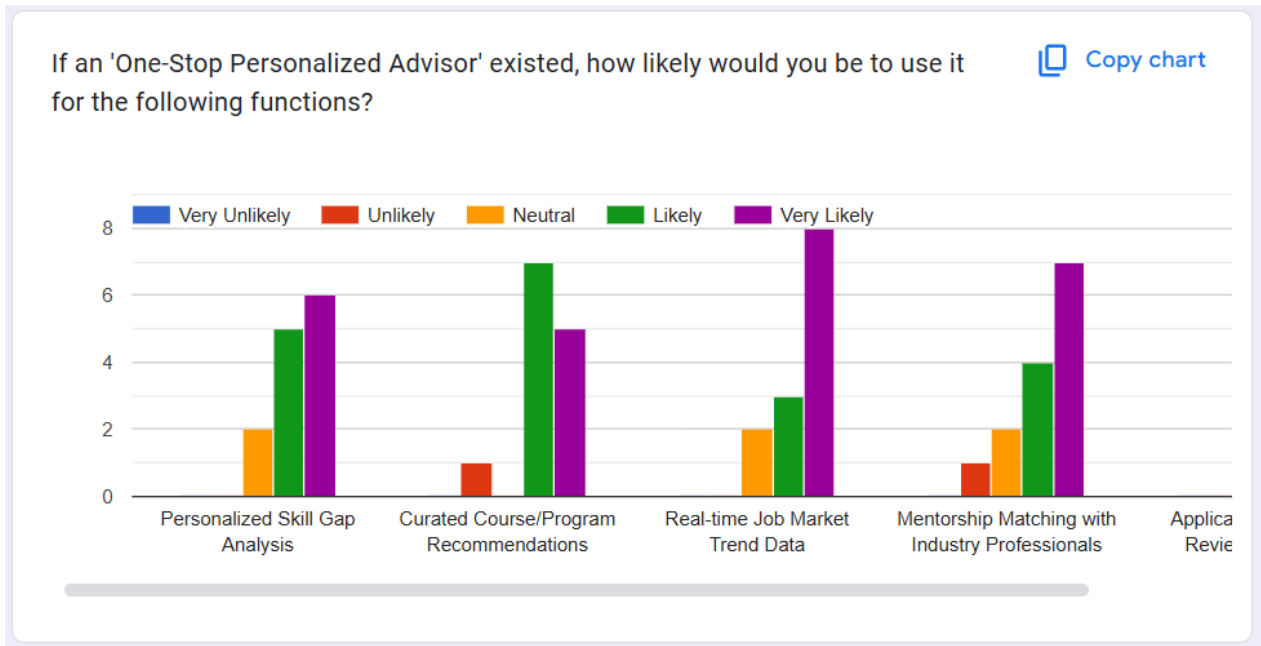


Figure 3.4: Likelihood of using features of a one-stop personalized advisor

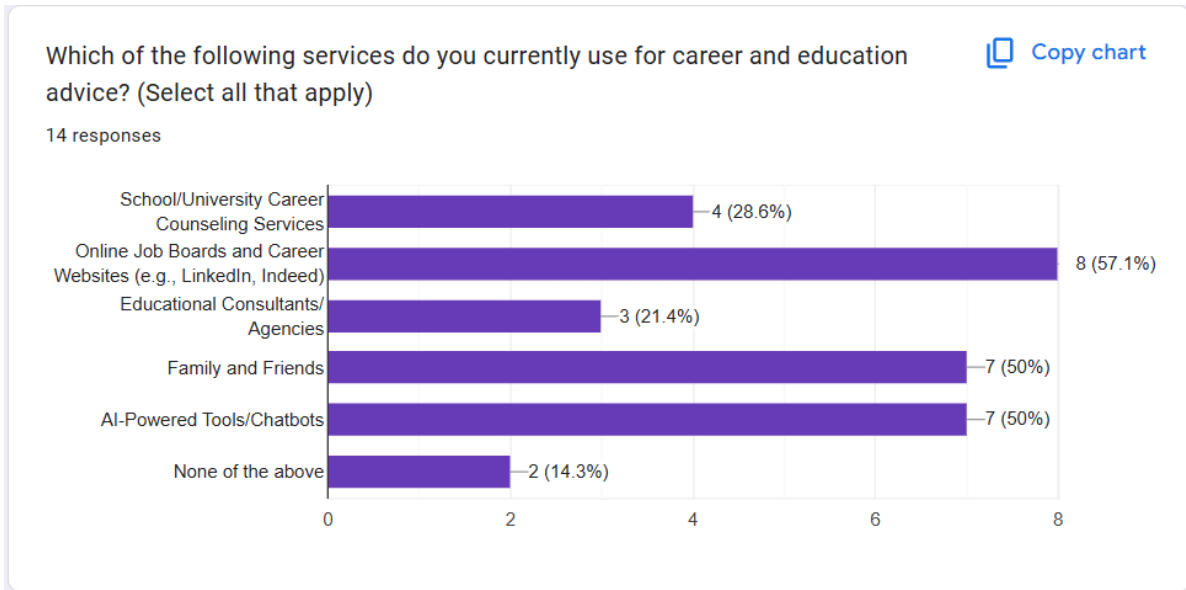


Figure 3.5: Current sources used for career and education advice

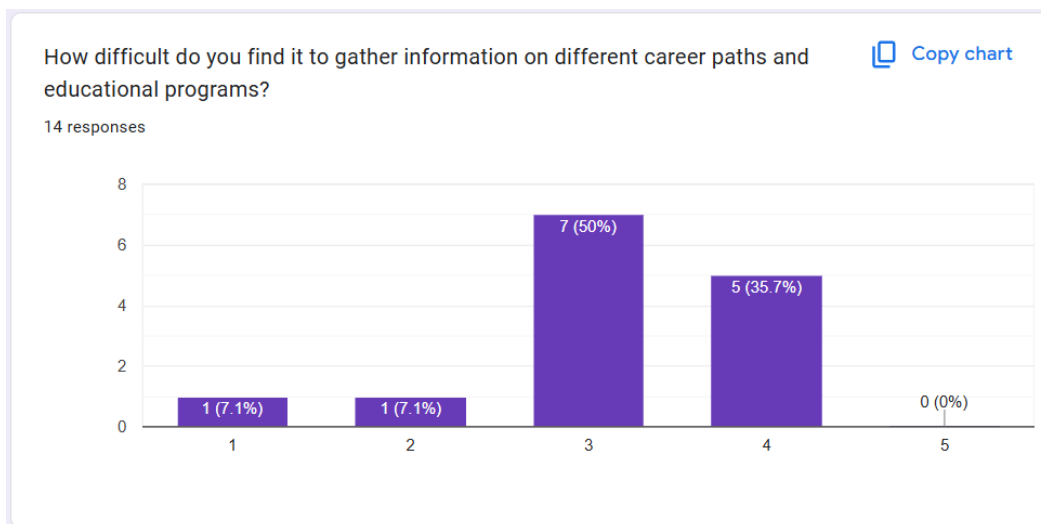


Figure 3.6: Difficulty level in gathering career and education information

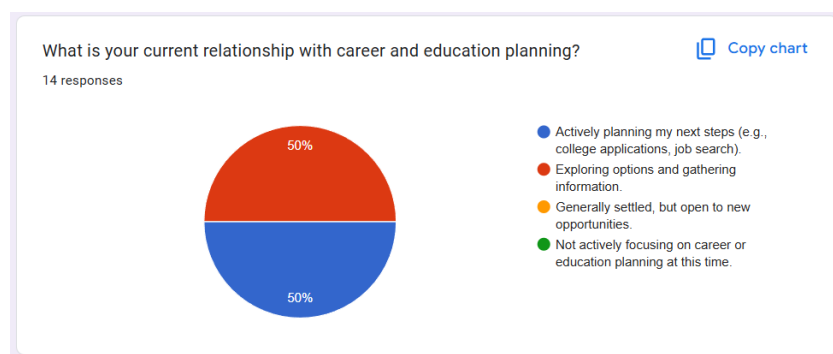


Figure 3.7: Current relationship of students with career and education planning

Chapter 4

Methodology

The methodology outlines the systematic approach adopted to design and develop the **One-Stop Personalized Career & Education Advisor**. The project is implemented in multiple phases to ensure accuracy, scalability, usability, and effectiveness in guiding students toward informed academic and career decisions. The overall approach follows best practices recommended for technology-driven educational guidance systems [2].

4.1 Requirement Analysis

In this phase, the functional and non-functional requirements of the system are identified by analyzing existing challenges faced by students after Class 10 and Class 12. The focus is on understanding student confusion regarding subject stream selection, availability of degree programs in government colleges, career outcomes, and admission timelines.

Inputs are gathered through the study of education policies, government college admission processes, and existing career guidance practices. National-level initiatives and education frameworks emphasize the need for structured and personalized guidance systems to support informed decision-making [1]. User roles such as students, parents, teachers, and administrators are defined, along with their interactions with the system. Key requirements include aptitude assessment, personalized recommendations, localized college information, and notification-based guidance.

4.2 System Design

The system is designed as a modular digital platform with a layered architecture consisting of a user interface layer, application logic layer, and data management layer. The user interface is designed to be simple and accessible through both web and mobile platforms to ensure ease of use for students from diverse backgrounds.

The application layer handles aptitude assessments, recommendation logic, course-to-career mapping, and timeline tracking. The backend data layer stores user profiles, assessment results, college information, course details, and career pathways. Secure APIs are used to enable data retrieval, personalization, and future integration with external educational portals, as recommended in modern digital education systems [3].

4.3 Content and Data Preparation

Educational content related to academic streams, degree programs, career opportunities, government examinations, and higher education pathways is collected from verified educational and government sources. The content is categorized into streams, courses, career outcomes, and eligibility requirements to ensure clarity and consistency.

Location-based data on government colleges, including available programs, facilities, eligibility criteria, and admission timelines, is organized and structured. This data is standardized to ensure accuracy across recommendations and visual mappings, addressing gaps identified in existing career guidance platforms.

4.4 Aptitude Assessment and Recommendation Development

Aptitude and interest-based assessments are designed using short quizzes to evaluate students' strengths, preferences, and personality traits. Such assessment-based guidance approaches have been shown to improve academic alignment and reduce indecision among students [4].

Recommendation logic is implemented to match student profiles with suitable academic paths and career options. Comparative views are provided to allow students to explore multiple stream and course combinations and understand their long-term implications.

4.5 Course-to-Career Mapping and Visualization

Course-to-career mapping modules are developed to visually represent the relationship between degree programs and career opportunities. These mappings include information on industries, job roles, government exams, higher studies, and entrepreneurial options associated with each course.

Interactive charts and structured pathways help students clearly understand how academic choices translate into real-world career outcomes. Visual representations are used to simplify complex information and enhance decision-making clarity.

4.6 Timeline Tracking and Notification System

A timeline tracking module is implemented to manage important academic events such as admission dates, entrance examinations, counseling schedules, and scholarship deadlines. Personalized notifications are generated based on user profiles and selected preferences.

This feature ensures that students remain informed and do not miss critical opportunities due to lack of awareness or delayed information, which is a commonly identified issue in higher education access [1].

4.7 Testing and Validation

The system undergoes comprehensive testing to evaluate functionality, accuracy of recommendations, usability, and performance. Test cases are executed across different user roles to validate assessment results, recommendation outputs, and data retrieval.

User feedback is collected during testing phases to refine the assessment logic, improve content clarity, and enhance the overall user experience. The objective is to ensure reliable guidance and ease of navigation.

4.8 Deployment

The finalized platform is deployed using a scalable cloud-based infrastructure to ensure availability and performance. Role-based authentication and data security measures are implemented to protect user information.

Regular updates to educational content, college data, and system features are planned to maintain relevance and accuracy. The deployed system enables continuous guidance and supports students throughout their academic decision-making journey.

Chapter 5

References

The following references have been consulted during the preparation of this project and the study of career guidance systems, educational policies, digital learning platforms, and technology-driven decision support systems:

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